MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON
TYPE 8759

The complete requirements for procuring the product described herein shall consist of this document and the issue in effect of Specification MIL-E-1.

DESCRIPTION: Integral cavity, reflex oscillator, 15.8 to 16.2 GHz.

OUTLINE AND DIMENSIONS: See Figure 1 herein.

ABSOLUTE RATINGS: (Note 1)

Parameters:	Ef	Ek	Ik	Er	F	TE	Altitude	Applied Tuner Torque
Units:	V	Vdc	mAdc	Vdc	GHz	°C	Feet	In-oz.
Maximum	6.9	350	45	-500	--	+150	55,000	100
Minimum:	5.7	--	--	-20	--	--	--	--
TEST CONDITIONS:								
Test Cond.(1):	6.3	300	--	-85 to -120	16.0	--	--	--
Test Cond.(2):	6.3	300	--	-140 to -200	16.0	--	--	--
Test Cond.(3):	6.3	--	--	--	--	--	--	--

MOUNTING: Output Flange, any position.

VOLTAGE REFERENCE: Cathode

FSC 5960

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PLUG CONNECTIONS: Notes 25, 26

GROUND: Anode grounded internally to tube shell.

COOLING: Forced air, convection, or heat sink sufficient to maintain envelope temperature below maximum. Monitor temperature at point A, as shown on Figure 1 and Note 21.

TUNING: Single shaft, clockwise rotation decreases frequency. (Note 35)

Method or Ref. Par.	Requirement or Test	Conditions (MIL-STD-1311 Par. 4)	AQL (% def.)	Insp. Level or Code	Symbol	Limits Min.	Limits Max.	Unit
	GENERAL							
---	Preproduction Sample Approval	Required (Note 23)	---	---	---	---	---	---
3.1	Performance	(Note 33)	---	---	---	---	---	---
1155	Cooling	(Note 21)	---	---	---	---	---	---
4.8.5	Holding period	t=168 hrs.	---	---	---	---	---	---
D-30	Dimensions	(Fig. 1)	---	---	---	---	---	---
3.6	Marking	(Note 34)						
	PREPRODUCTION SAMPLE APPROVAL TESTS, (Note 23)							
---	Low Pressure	Test Condition(1) Notes 6, 7, 9			F	--	17	MHz
---	Vibration(1)	Test Condition(1) Notes 7, 10			Ir	--	10	uAdc
---	Vibration(2)	Test Condition(1) Notes 7, 11			ΔF (p-p)	--	5	MHz
					$\Delta Po/Po$	--	10	%
---	Temperature Coefficient	Test Condition(1) Notes 2, 6, 7			$\Delta F/\Delta T$	--	0.5	MHz/°C
4223	Mechanical Tuning Fatigue	Test Condition(1) Note 3			---	1000	--	Cycles

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PREPRODUCTION SAMPLE APPROVAL TESTS, (Note 23) (Cont'd)

---	Shock	Test Condition(1) Notes 7, 8	F	--± 12		MHz
1011	Humidity	Note 32	---	--	--	---

Method or Ref.	Par.	Requirement or Test	Conditions (MIL-STD-1311 Par. 4)	AQL (% def.)	Insp. Level or Code	Symbol	Limits Min.	Limits Max.	Unit
			QUALITY CONFORMANCE INSPECTION, PART 1 Notes 27, 28						
---		Continuity	Note 12		---	---	---		---
---		Shorts	Note 13		---	---	---		---
1301		Heater Current	Ef=6.3V		If	500	600		mA
4229		Total Reflector Current	Test Condition(2) Notes 4, 14		Irt	--	10		uAdc
4229		Reflector Leakage Current	Test Condition(2) Notes 4, 14		Irl	--	6		uAdc
4229		Reflector Gas Current	Test Condition(2) Notes 4, 14		Irg	--	4		uAdc
1256		Cathode Current	Test Condition(1) Note 7		Ik	--	35		mAdc
4214		Cathode Emission	Test Condition(1) Notes 5, 7, 14		Ik	--	10		%
4250		Power Output(1)	Test Condition(1)						
			F=15.8 GHz		Po	20	--		mW
			F=16.2 GHz		Po	20	--		mW
			Notes 6, 7						

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Method or Ref. Par.	Requirement or Test	Conditions MIL-STD-STD-1311 Par.4)	AQL (1% def.)	Insp. Level or Code	Symbol	Limits Min.	Limits Max.	Unit
		QUALITY CONFORMANCE INSPECTION, PART 1 Notes 27, 28 (Cont'd)						
4213	Reflector Voltage	Test Condition(1) F=15.8 GHz F=16.2 GHz Notes 6, 7			Er	-100 --	-- -120	Vdc Vdc
4231	Electronic Tuning Hysteresis	Test Condition(1) F=15.8 to 16.2 GHz (inclusive) Notes 6, 7, 15			h	--	50	%
---	Mode Centering	Test Condition(1) Er=-110 Vdc F=16.0 GHz Notes 6, 22			Po	---	0.2	db
		QUALITY CONFORMANCE INSPECTION, PART 2						
---	Electrode Insulation	Note 16	6.5	L6	Ik-rs ih-rs	-- --	150 150	uAdc uAdc
---	Mode Continuity	Test Condition Notes 7, 17	6.5	L6	---	--	---	---
4200	Electronic Tuning Range	Test Condition(1) 1/2 Power Notes 6, 7	6.5	L6	F	50	--	Mc
---	Reflector Voltage For Electronic Tuning	Test Condition(1) F=15.8 GHz F=16.2 GHz Notes 6, 7, 18	6.5	L6	Er Er	-- --	35 35	Vdc Vdc
---	Modulation Sensitivity	Test Condition(1) Notes 6, 7, 19	6.5	L6		1.3	3.5	MHz/v
---	Tuner Torque	Note 29	6.5	L6		--	32	in-Oz
---	Mechanical Tuning Range	Test Condition(1) Notes 7, 20	6.5	L6	---	--	---	---

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Method or Ref.Par.	Requirement or Test	Conditions (MIL-STD-1311 Par. 4)	AQL (1% def.)	Insp. Level or Code	Symbol	Limits Min.	Limits Max.	Unit
		QUALITY CONFORMANCE INSPECTION, PART 2 (Cont'd)						
4211	Power Output(2)	Test Condition(1) ER=-110Vdc Note 6	6.5	L6	Po	20	--	mW
---	Tuner Shaft Sensitivity(1)	Test Condition(1) Note 7	6.5	L6	---	--	350	MHz/turn
---	Tuner Shaft Sensitivity(2)	Test Condition(1) Notes 7, 31	6.5	L6	Er	--	8	V/turn
---	Tuner Stop Torque Test	Torque=100 in.-oz max. Note 30	6.5	L6	---	--	--	oz. in.
		QUALITY CONFORMANCE INSPECTION, PART 3, LIFE						
4.6	Life Test	Test Condition(1) Group 9	---	--	--	--	--	
1501	Intermittent Life Test Operational	Notes 7, 21, 36 2500 hours						
4.6.2	Life Test End Point Power Output(1)	--	--		Po	15	--	mW
	Rough-handling test	Per MIL-E-75 or contract						
1136	Rough-handling Test End Points		--	--	--	--	--	
5.1	Preparation for Delivery	As specified in contract or order	--	--	--	--	--	

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NOTES

Note 1: Refer to Specification MIL-E-1, Section 6 "Note" covering absolute maximum ratings.

Note 2: Temperature Coefficient Test Procedure:

(a) Secure tube firmly with metal to metal contact to a section of standard waveguide at least 24 inches in length, ten (10) inches of which shall be included in the temperature chamber.

(b) Close oven door and apply test voltages to the tube. Set the output frequency of the tube to approximately 16.0 GHz.

(c) Allow the tube to stabilize for approximately 30 minutes in still air at room temperature. At the end of the 30 minutes adjust the reflector voltage to maximum power on Mode A and record frequency and temperature. Do not readjust the reflector voltage at any time during the remainder of the test.

(d) Adjust the oven to obtain an ambient temperature of approximately 105°C. Allow the tube to stabilize for 30 minutes in still air and record the frequency and the temperature.

(e) The temperature coefficient is the difference between the two frequency readings divided by the difference between the two temperature readings.

Note 3: After the specified number of cycles of Mechanical Fatigue Test starting at 15.8 GHz to 16.2 GHz and return to 15.8 GHz the tube shall pass the Quality Conformance Inspection - Part 1 requirement of the test specification.

Note 4: Apply heater voltage for one minute. Then apply other voltages plus the heater voltage for two minutes, and read the total reflector current at the end of this two minute period.

Note 5: Reduce heater voltage from 6.3 volts to 5.7 volts. The percentage change in resonator current after one minute shall not exceed the specified value.

Note 6: All oscillation tests are to be made with the tube rigidly connected to a UG-419/U cover flange on appropriate RG-91/U waveguide. The RF load shall have a maximum VSWR of 1.1 to 1.

Note 7: The reflector voltage is to be adjusted within the specified range for maximum power on Mode A at the frequency specified. Mode A can be found between Er=-100V and -120V.

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Note 8: The tube is to be given 5 shocks at 70 g's and 6.5 milliseconds duration of sine pulse configuration in each of three mutually perpendicular planes. The tube must be mounted securely to the waveguide flange. The frequency shift shall be read following each shock, and the average shift resulting from five shocks shall not exceed the limit specified. The tube shall meet all other Quality Conformance Inspection - Part 1 test conditions after completion of shock testing.

Note 9: The frequency shall be stabilized at a pressure of 70 mm of Hg. The pressure shall be increased to 760 mm of Hg with uncontrolled temperature. The pressure change shall be complete and the frequency change measured within 10 seconds. The frequency change shall not exceed the limit specified. The tube is to be fastened to a waveguide which is maintained at atmospheric pressure during the test.

Note 10: The tube shall be vibrated at 10 g's at a frequency of 60 Hz for two minutes in three mutually perpendicular planes. The reflector current bursts shall be recorded with an instrument whose frequency response is from d.c. to 100 Hz with no more than 3 db fall-off at 100 Hz. There shall be no reflector current bursts greater than the limit specified.

Note 11: The tube shall be vibrated at 10 G's for five minutes in each of three mutually perpendicular planes over the frequency range of 40 to 500 Hz at logarithmic sweep, and shall require the time specified to traverse the range. The frequency modulation and the change in power output, resulting from vibration shall not exceed the limits specified. The tube shall meet all Quality Conformance Inspection - Part 1 tests following vibration. The total power output change shall not exceed 20%.

Note 12: Continuity shall exist between pins C and E and between pins A and C.

Note 13: Continuity shall not exist between any of the following:

A and H
C and D
D and H

Note 14: This test is to be performed after the holding period.

Note 15: The mode discontinuity percentage shall be defined as the ratio of the power level at which a discontinuity in the mode shape is exhibited to the maximum power level.

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Note 16: Apply 300 Vdc between the elements specified. The resulting current shall not exceed the limits specified. No other voltages are to be applied.

Note 17: With a 1.5 to 1 voltage standing wave inserted in the guide, there shall be no discontinuity at the maximum power points for any phase of standing wave.

Note 18: The change in reflector voltage required to tune across the specified minimum electronic tuning range (not to 1/2 power points) shall not exceed the limit specified. The center of the electronic tuning range is to be the maximum power point of the mode.

Note 19: Modulation Sensitivity is defined as defined as the specified Electronic Tuning Range (50 MHz) divided by the reflector voltage between the $\pm$25 MHz points.

Note 20: Mechanical Tuning Range: The total number of turns of the tuner shaft between mechanical stops shall be 6 to 9 turns. The midpoint of the total tuner shaft travel shall be the reference point for measurement of the tuning range. The specified tuning range of 15.8 to 16.2 GHz shall be covered in any portion of the travel of the tuning shaft within $\pm$2 turns of the reference point as limited by the Tuner Shaft Sensitivity (1) requirement.

Note 21: Tubes shall operate at a ambient temperature of 25° to 40°C with no cooling except convection and conduction to the wave guide (UG-419).

Note 22: With reflector voltage set at the specified value, the power output shall be within the amount specified of the power output obtainable by optimizing the reflector voltage at the frequency specified.

Note 23: Preproduction Sample Approval requirements hereby replace any Qualification requirements referable to the product covered herein. The term "First Article Testing" shall be considered as synonomous with the term "Preproduction Sample Approval Testing." All tests applicable herein (including all Preproduction Sample Approval and Quality Conformance Inspection, Parts 1, 2 and 3) shall be performed during Preproduction Sample Inspection. A failure of any one tube in any of the tests shall be cause for decision of Preproduction Sample disapproval.

Note 24: Wire shall be MIL-W-16878/7 Type F-22

Note 25: Connector Plug shall be Continental Type No. C7-20P-V SCLHC or Amphenol Type 126-195.

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Note 26: Plug Connections shall be as follows:

Element	Pin Connection
Cathode	A
Heater	C
Reflector	D
Heater	E
Shell	H

Note: Wire size described in Note 24

All testing of the tube to be done only after assembly of plug to the tube.

Note 27: The AQL for the combined defectives for attributes in Acceptance Inspection, Part 1, Production, excluding mechanical and inoperatives, shall be 1.0%.

Note 28: All tests on this specification shall be performed at the conclusion of the specified non-operating holding period commencing with the total reflector current test.

Note 29: Under the conditions specified, the tuning mechanism shall be smoothly operable throughout the entire mechanical tuning range without the specified torque being exceeded.

Note 30: The tuning mechanism shall withstand the maximum specified stop torque at the extreme ends of the tuning range without damage.

Note 31: Tune tube to 15.8 GHz and measure the reflector voltage at mode center. Turn tuner one turn counter clockwise and measure reflector voltage at mode center for this tuner position. The difference in reflector voltage divided by the one turn change in tuner position shall not exceed the limit specified.

Note 32: After completion of this test, tube shall meet requirements of Power output (1) and tuner torque.

Note 33: Associable with and in addition to the performance requirements covered herein, all tests listed in Specification MIL-E-1, paragraph 3.1, are applicable except those per (therein-) referred MIL-STD-1311 Methods E-1011, E-1111 and E-1201.

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Note 34: Marking shall be in accordance with Specification MIL-E-1. If any specification-requirements waiver has been granted, the product-identification marking shall consist of the tube type-number only.

Note 35: There shall be no lateral movement of the tuning shaft as it is rotated through the mechanical tuning range.

Note 36: Life-test data and curves of failure are also to be submitted to the following in addition to the requirement of MIL-E-1 para. 4.6.1.9.3.1:

Commanding General
US Army Electronics Command
ATTN: AMSEL-PP-EM-1
Fort Monmouth, New Jersey 07703

CUSTODIAN

ARMY EL

Preparing Activity

ARMY EL

Project No. 5960-A846

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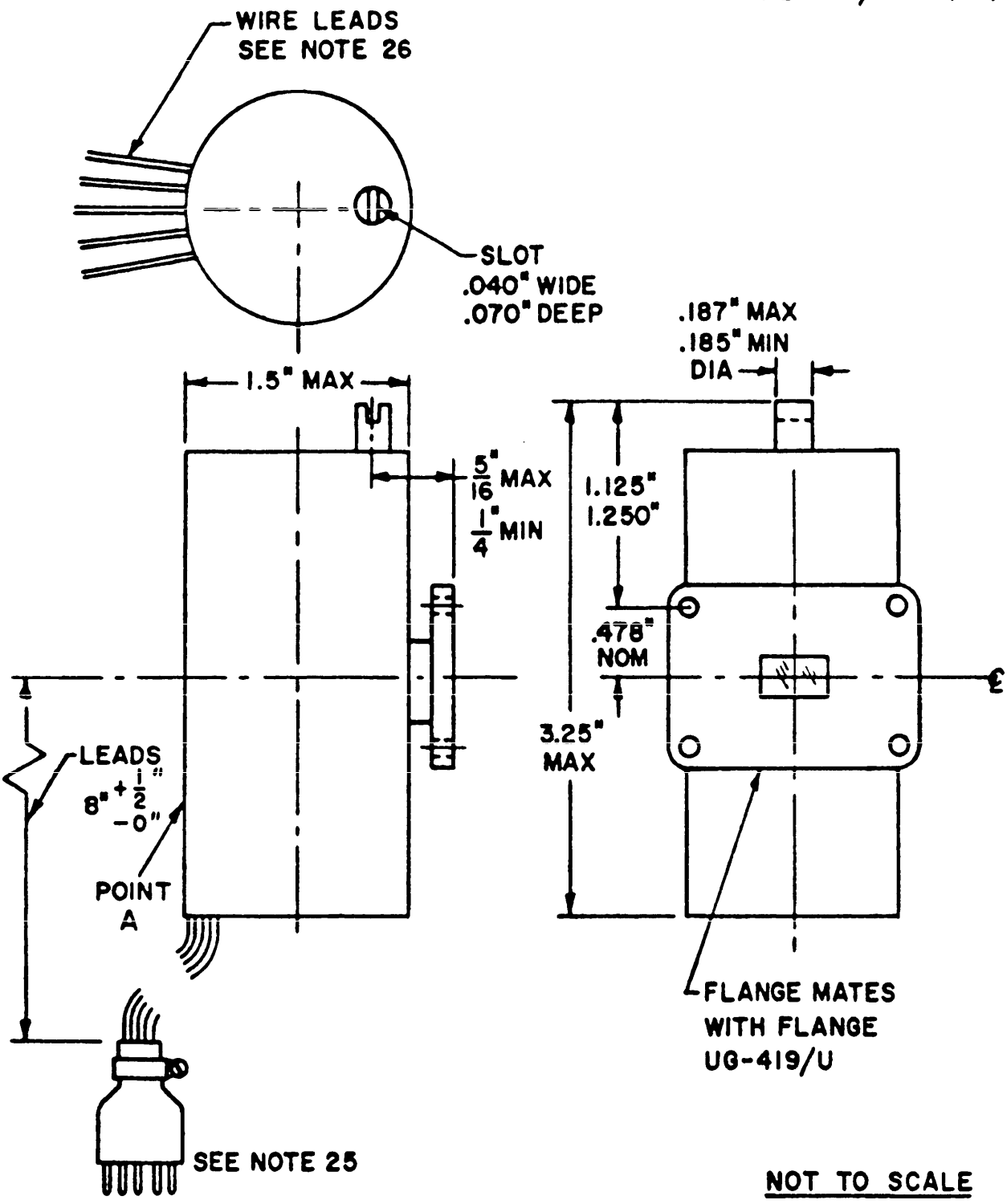


FIGURE 1. OUTLINE AND DIMENSIONS

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-E-1/1677(EL), ELECTRON TUBE, KLYSTRON TYPE 8759

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-PM 1068-68

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